# Korean Stroke Registry
# Annual Report
# *2020*

Korean Stroke Registry
한국뇌졸중등록사업

# KSR Annual Report – **2020**

Korean Stroke Registry

# About Korean Stroke Registry (KSR)

Korean Stroke Registry(KSR) is a representative multi-center, prospective, and hospital-based stroke registry in Korea. KSR established to collect data on patients with acute stroke and transient ischemic attack, and to monitor, evaluate, and improve the quality of stroke care in Korea.

In 2001, KSR was launched with 33 major hospitals from 23 medical schools with support from the Korean Stroke Society. In 2006, KSR was progressed with the participation of 66 major hospitals, as one of the six divisions of the Clinical Research Center for Stroke (CRCS) with the support of the Ministry of Health and Welfare, Republic of Korea.

In 2016, the Korean Stroke Society established the KSR steering committee and is operating a new KSR system with the support of the Korea Disease Control and Prevention Agency, Republic of Korea.

As of 2020, 79 hospitals nationwide are participating in the KSR, and 47 hospitals are participating in the KSR Core DB, and participating hospitals are gradually expanding.

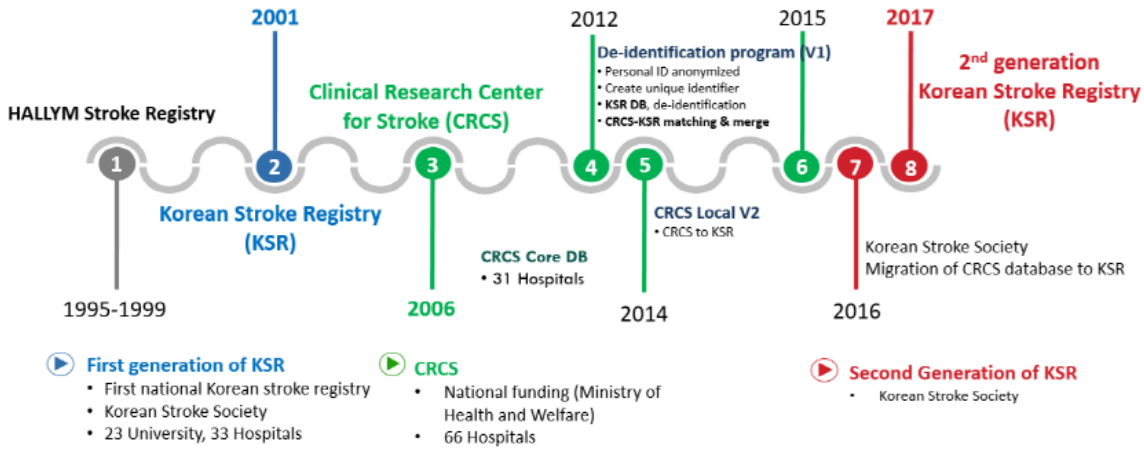


Figure 1. Historical Review of Korean Stroke Registry

# Goals of KSR

The overall goal of the KSR is to provide reliable and representative national data on acute stroke care and outcome of patients with all stroke or transient ischemic attack cases in Korea

- To assess the characteristics and secular trend of acute stroke in Korea
- To monitor the quality of acute stroke care
- To implement professional guidelines for acute stroke care
- To monitor the impact of care on post discharge patient outcomes
- To develop a large comprehensive data set for stroke research

# Database of KSR

The database of KSR composes a comprehensive and integrated patient management system by supplementing the problems of existing disease-specific and research-specific clinical databases. Reflecting the medical reality, the Core DB was configured to allow input of common data in multiple medical institutions, and the Extended DB was configured to allow data to be added in each study. It is also configured to add long-term prognosis and functional status information (Outcome DB).

In order to manage the data quality of KSR, data are backed up in real-time, and an error check system is automatically applied to reduce errors that may occur during the data entry process. All personal information data has been completely removed through a de-identification program. In addition, the quality of data is improved by managing missing and abnormal input values as a query through the quality management of the database.

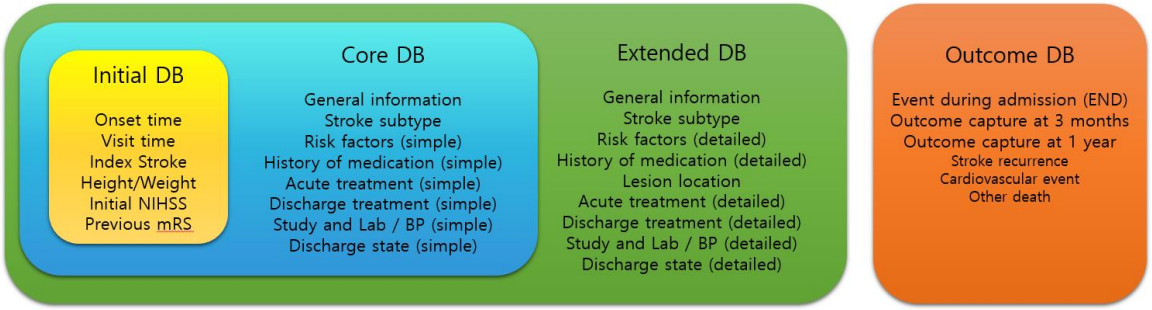


Figure 2. Database structure of Korean Stroke Registry

# Participating stroke centers

(KSR Core DB, 47 hospitals, alphabetical order)

The KSR annual report 2020 analyzed the cases of 38 hospitals (marked with an asterisk) that continuously participated in the audit process from Jan 2012 to Oct 2019.

- Asan Medical Center
- Bundang Jesaeng Hospital*
- Changwon Fatima Hospital*
- Chonnam National University Hospital*
- Chung-Ang University Hospital
- Chungbuk National University Hospital*
- Chungnam National University Sejong Hospital
- Daegu Caholic University Medical Center*
- Daejeon Eulji Medical Center, Eulji University*
- Dong-A University Hospital*
- Dongguk University Ilsan Hospital*
- Nowon Eulji Medical Center, Eulji University*
- Gachon University Gil Medical Center*
- Gangnam Severance Hospital*
- Gyeonggi Provincial Medical Center Icheon Hospital
- Gyeongsang National University Hospital*
- Gyeongsang National Universtiy Changwon Hospital*
- Hallym University Chuncheon Sacred Heart Hospital*
- Hallym University Dongtan Sacred Heart Hospital*
- Hallym University Sacred Heart Hospital*

- Hanyang University Hospital*
- Incheon Paik Hospital
- Inha University Hospital*
- Inje University Busan Paik Hospital*
- Inje University Ilsan Paik Hospital*
- Jeju National University Hospital*
- Kangdong Sacred Heart Hospital*
- Kangwon National University Hospital*
- Keimyung University Dongsan Hospital*
- Konkuk University Medical Center*
- Korea University Guro Hospital*
- Kosin University Gospel Hospital*
- Kyung Hee University Medical Center*
- Kyungpook National University Chilgok Hospital*
- Mokpo Jung-Ang Hospital
- Pohang Stroke and Spine Hospital*
- Samsung Medical Center*
- Seoul Medical Center*
- Seoul National University Bundang Hospital*
- Seoul National University Hospital*
- Severance Hospital
- SMG-SNU Boramae Medical Center*
- Soonchunhyang University Hospital*
- Ulsan University Hospital*
- Yeungnam University Medical Center*
- Yonsei S Hospital
- Yuseong Sun Hospital

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# Abbreviations

| | |
|---|---|
| BMI | Body mass index |
| DBP | Diastolic blood pressure |
| HDL | High-density lipoprotein |
| IQR | Interquartile range |
| LDL | Low-density lipoprotein |
| mRS | modified Rankin Scale |
| NIHSS | National Institutes of Health Stroke Scale |
| SBP | Systolic blood pressure |
| SD | Standard deviation |
| TG | Triglyceride |
| TOAST | Trial of ORG 10172 in Aacute Stroke Treatment |

# Global Statistics of KSR

*Demographic profile of acute ischemic stroke*

- Of 100,914 stroke cases registered in 38 hospitals between Jan 2012 and Oct 2019, number of acute ischemic stroke was 89,277 (88.5%).
- Men were 52,375 (59%) and women were 36,902 (41%).
- Mean (SD) age at onset was 65.7 (12.6) for men and 71.9 (12.7) for women.

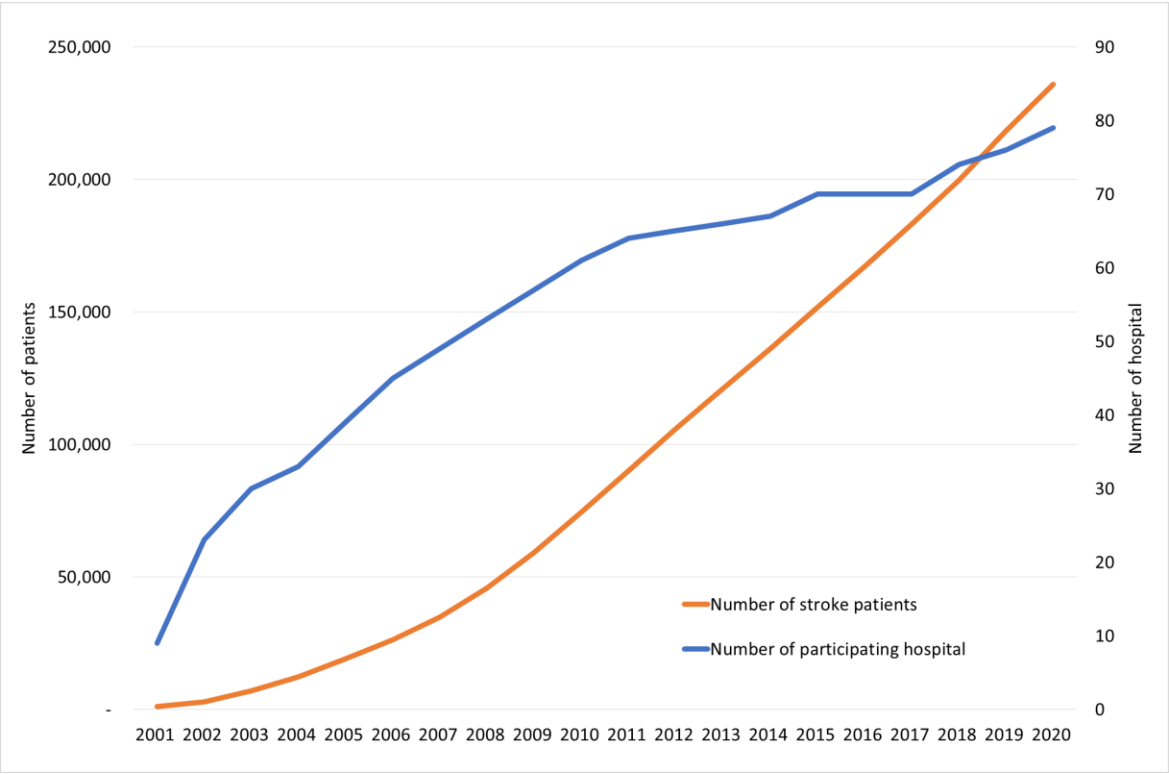


Figure 3. Number of stroke patients and participating hospital since 2001

All cases registered in KSR, including non-audited cases

Table 1. Number of registered acute stroke patients

| | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | Sum |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2012 | 868 (8.8) | 813 (8.2) | 894 (9.1) | 757 (7.7) | 828 (8.4) | 859 (8.7) | 868 (8.8) | 778 (7.9) | 787 (8.0) | 846 (8.6) | 831 (8.4) | 746 (7.6) | 9875 |
| 2013 | 891 (8.3) | 779 (7.3) | 915 (8.6) | 910 (8.5) | 916 (8.6) | 831 (7.8) | 912 (8.5) | 935 (8.8) | 856 (8.0) | 884 (8.3) | 923 (8.6) | 923 (8.6) | 10675 |
| 2014 | 906 ( 8.4) | 774 (7.2) | 861 (8.0) | 857 (8.0) | 927 (8.6) | 837 (7.8) | 856 (8.0) | 868 (8.1) | 932 (8.7) | 918 (8.5) | 977 (9.1) | 1034 (9.6) | 10747 |
| 2015 | 917 (7.2) | 974 (7.7) | 1048 (8.3) | 1053 (8.3) | 1133 (8.9) | 1032 (8.1) | 1117 (8.8) | 1083 (8.5) | 1069 (8.4) | 1052 (8.3) | 1112 (8.8) | 1084 (8.6) | 12674 |
| 2016 | 1085 (8.3) | 1030 (7.9) | 1102 (8.4) | 1070 (8.2) | 1113 (8.5) | 1075 (8.2) | 1071 (8.2) | 1076 (8.2) | 1061 (8.1) | 1227 (9.4) | 1106 (8.4) | 1094 (8.3) | 13110 |
| 2017 | 1059 (7.5) | 970 (6.8) | 1077 (7.6) | 1187 (8.4) | 1171 (8.3) | 1234 (8.7) | 1253 (8.8) | 1263 (8.9) | 1229 (8.7) | 1208 (8.5) | 1222 (8.6) | 1320 (9.3) | 14193 |
| 2018 | 1328 (8.5) | 1112 (7.1) | 1202 (7.7) | 1232 (7.9) | 1317 (8.4) | 1290 (8.3) | 1470 (9.4) | 1318 (8.4) | 1217 (7.8) | 1417 (9.1) | 1308 (8.4) | 1405 (9.0) | 15616 |
| 2019 | 1408 (10.0) | 1207 (8.6) | 1348 (9.6) | 1393 (9.9) | 1477 (10.5) | 1410 (10.1) | 1462 (10.4) | 1446 (10.3) | 1374 (9.8) | 1499 (10.7) | | | 14024 |
| Sum | 8462 (8.4) | 7659 (7.6) | 8447 (8.4) | 8459 (8.4) | 8882 (8.8) | 8568 (8.5) | 9009 (8.9) | 8767 (8.7) | 8525 (8.5) | 9051 (9.0) | 7479 (7.4) | 7606 (7.5) | 100914 |

Values are presented as number (row percent)

All audited cases registered in KSR

Table 2. Number of acute ischemic stroke patients stratified by sex and age group

| Age group | All (n) | Per (%) | Male(n) | Per (%) | Female(n) | Per (%) |
|---|---|---|---|---|---|---|
| <45 | 4227 | 4.7 | 2884 | 5.5 | 1343 | 3.6 |
| 45-54 | 9503 | 10.6 | 7002 | 13.4 | 2501 | 6.8 |
| 55-64 | 17964 | 20.1 | 13133 | 25.1 | 4831 | 13.1 |
| 65-74 | 24571 | 27.5 | 14827 | 28.3 | 9744 | 26.4 |
| 75-84 | 26114 | 29.3 | 12315 | 23.5 | 13799 | 37.4 |
| ≥85 | 6898 | 7.7 | 2214 | 4.2 | 4684 | 12.7 |
| Mean (SD) | 68.3 (13.0) | | 65.7 (12.6) | | 71.9 (12.7) | |
| Total | 89277 | 100 | 52375 | 100 | 36902 | 100 |

Based on 89,277 ischemic stroke cases

Table 3. Secular trends of age group proportions in acute ischemic stroke

| Age group, y | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|
| <45 | 4.9 | 5.1 | 5.2 | 4.6 | 4.8 | 4.8 | 4.5 | 4.4 |
| 45-54 | 11.5 | 11.9 | 11.1 | 10.7 | 10.6 | 10.1 | 10.1 | 9.9 |
| 55-64 | 19.2 | 19.7 | 19.6 | 19.8 | 20.6 | 20.2 | 20.4 | 20.9 |
| 65-74 | 31.4 | 30.0 | 29.0 | 28.5 | 27.2 | 26.2 | 25.4 | 25.1 |
| 75-84 | 26.8 | 27.2 | 27.8 | 29.5 | 29.6 | 30.6 | 30.6 | 30.1 |
| ≥85 | 6.3 | 6.2 | 7.3 | 6.8 | 7.4 | 8.0 | 8.9 | 9.6 |

Based on 89,277 ischemic stroke cases

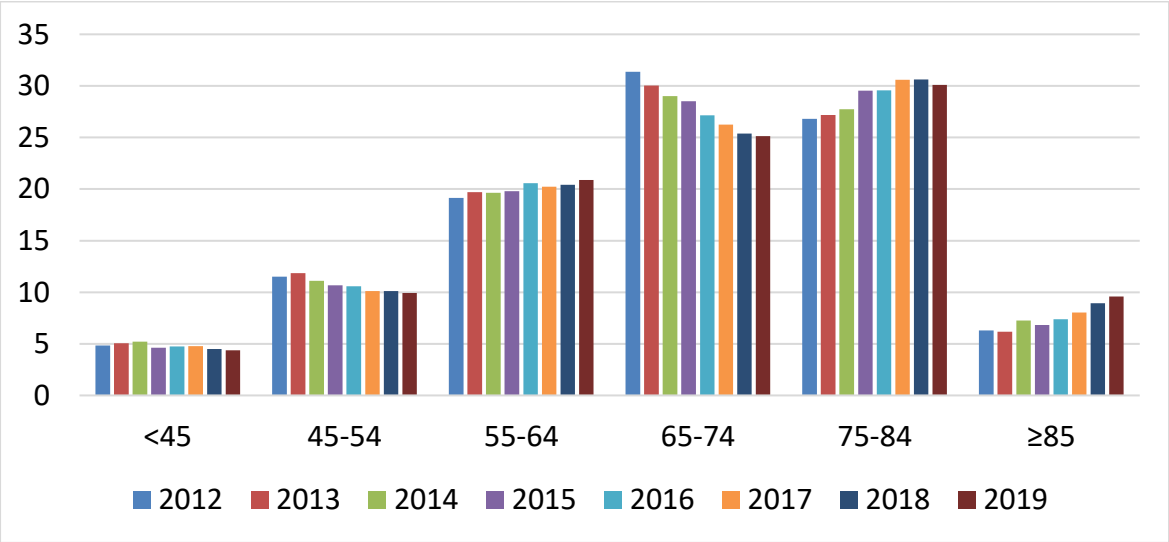


Figure 4. Secular trends of age group proportions in acute ischemic stroke

Table 4. Frequency of TOAST classification subtypes

| | Frequency (n) | Percentage (%) |
|---|---|---|
| Large Artery Atherosclerosis | 29308 | 32.83 |
| Small Vessel Occlusion | 17829 | 19.97 |
| Cardioembolism | 18330 | 20.53 |
| Other Determined Etiology | 3574 | 4.00 |
| Undetermined Etiology - 2 or more | 4480 | 5.02 |
| Undetermined Etiology - Negative | 8229 | 9.22 |
| Undetermined Etiology - Incomplete work-ups | 6596 | 7.39 |
| Missing | 931 | 1.04 |

Based on 89,277 ischemic stroke cases

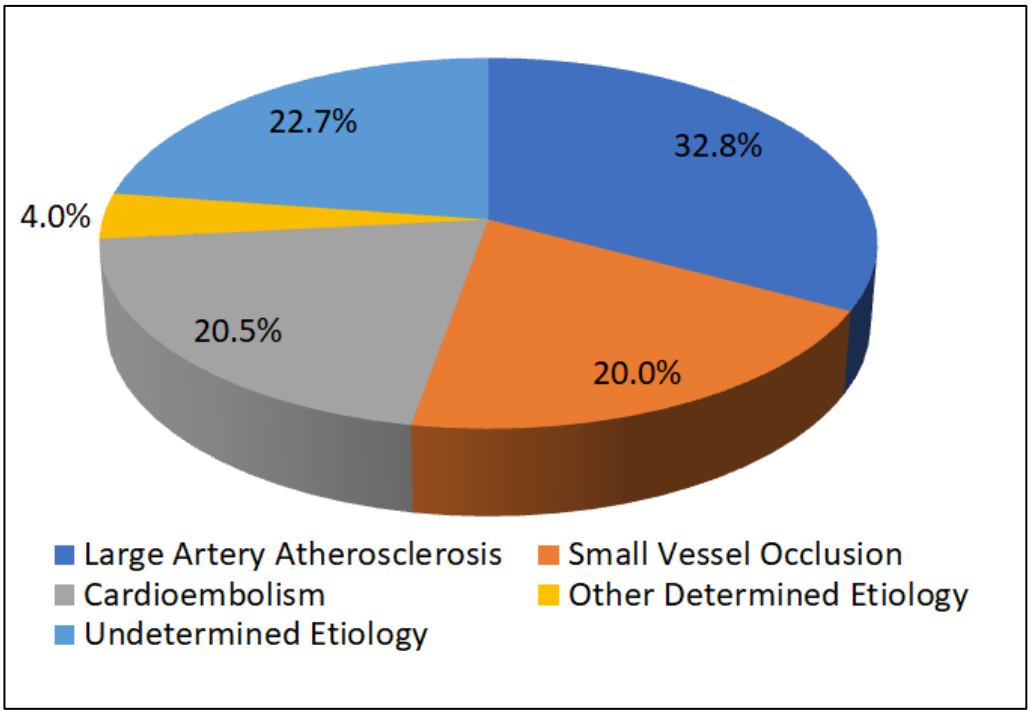


Figure 5. TOAST classification

Based on 89,277 ischemic stroke cases

Table 5. Secular trends of TOAST Classification

| | Number of patients (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
| Large Artery Atherosclerosis | 3107 (37.3) | 3089 (34.3) | 3138 (33.3) | 3509 (32.3) | 3750 (33.0) | 4069 (32.7) | 4565 (32.3) | 4081 (31.9) |
| Small Vessel Occlusion | 1582 (19.0) | 1838 (20.4) | 1766 (18.7) | 2139 (19.7) | 2275 (20.0) | 2592 (20.8) | 2951 (20.9) | 2686 (21.0) |
| Cardioembolism | 1782 (21.4) | 1918 (21.3) | 2089 (22.2) | 2360 (21.7) | 2305 (20.3) | 2459 (19.8) | 2874 (20.3) | 2543 (19.9) |
| Other Determined Etiology | 226 (2.7) | 342 (3.8) | 354 (3.8) | 367 (3.4) | 412 (3.6) | 509 (4.1) | 779 (5.5) | 585 (4.6) |
| Undetermined Etiology - 2 or more | 427 (5.1) | 410 (4.6) | 478 (5.1) | 608 (5.6) | 595 (5.2) | 659 (5.3) | 683 (4.8) | 620 (4.9) |
| Undetermined Etiology - Negative | 550 (6.6) | 659 (7.3) | 752 (8.0) | 1043 (9.6) | 1141 (10.0) | 1342 (10.8) | 1387 (9.8) | 1355 (10.6) |
| Undetermined Etiology - Incomplete work-ups | 662 (7.9) | 746 (8.3) | 848 (9.0) | 848 (7.8) | 882 (7.8) | 804 (6.5) | 898 (6.4) | 908 (7.1) |
| Total | 8336 | 9002 | 9425 | 10874 | 11360 | 12434 | 14137 | 12778 |

Based on 88,346 ischemic stroke cases

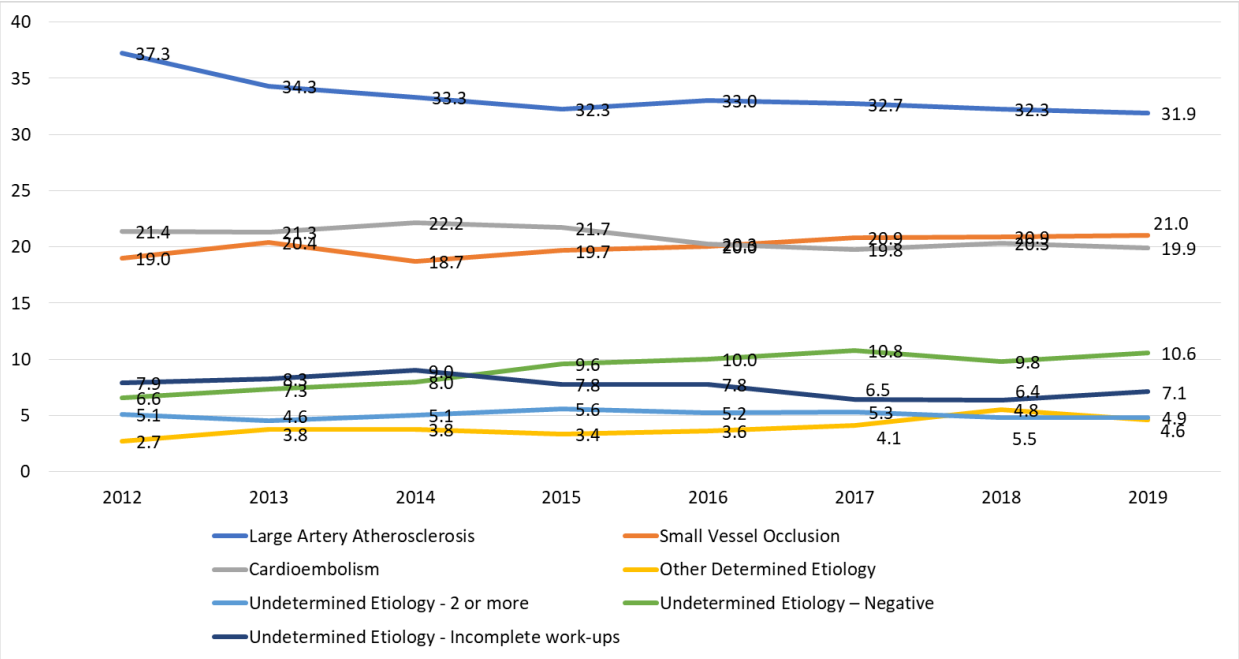


Figure 6. Secular trends of proportions in TOAST Classification

Table 6. Time interval between symptom onset and hospital arrival

| | Number of patients (%) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Seoul | Gyeonggi & Incheon | Gyeongnam & Busan | Chungnam & Daejeon | Gyeongbuk & Daegu | Gangwon | Gwangju & Jeonnam | Chungbuk | Jeju |
| ≤3 hour | 5726 (22.2) | 5925 (24.8) | 5207 (27.8) | 1036 (27.8) | 3032 (23.1) | 609 (26.6) | 2311 (27.5) | 730 (31.7) | 701 (30.0) |
| 3-6 hour | 2387 (9.3) | 2425 (10.2) | 2446 (13.0) | 357 (9.6) | 1494 (11.4) | 249 (10.9) | 1445 (17.2) | 244 (10.6) | 268 (11.5) |
| 6-12 hour | 3378 (13.1) | 3113 (13.0) | 2736 (14.6) | 525 (14.1) | 1763 (13.4) | 289 (12.6) | 1287 (15.3) | 326 (14.2) | 346 (14.8) |
| 12-24 hour | 4436 (17.2) | 4071 (17.0) | 3177 (16.9) | 650 (17.4) | 2370 (18.1) | 405 (17.7) | 1212 (14.4) | 414 (18.0) | 392 (16.8) |
| 24-36 hour | 2044 (7.9) | 1706 (7.1) | 1219 (6.5) | 302 (8.1) | 932 (7.1) | 165 (7.2) | 711 (8.5) | 116 (5.0) | 150 (6.4) |
| 36-48 hour | 1508 (5.9) | 1225 (5.1) | 749 (4.0) | 188 (5.0) | 710 (5.4) | 109 (4.8) | 237 (2.8) | 143 (6.2) | 81 (3.5) |
| 2-7 day | 5352 (20.8) | 4476 (18.7) | 2725 (14.5) | 607 (16.3) | 2331 (17.8) | 390 (17.0) | 1021 (12.1) | 323 (14.0) | 365 (15.6) |
| >7 day | 936 (3.6) | 942 (3.9) | 500 (2.7) | 61 (1.6) | 479 (3.7) | 76 (3.3) | 190 (2.3) | 5 (0.2) | 30 (1.3) |
| **TOTAL** | 25767 | 23883 | 18759 | 3726 | 13111 | 2292 | 8414 | 2301 | 2333 |

Based on 100,586 acute stroke patients

Table 7. Secular trends of categorized symptom onset to arrival time

| | Number of patients (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
| ≤3 hour | 2364 (24.1) | 2594 (24.5) | 2748 (25.7) | 3136 (24.8) | 3361 (25.7) | 3552 (25.1) | 3930 (25.2) | 3592 (25.6) |
| 3-6 hour | 1197 (12.2) | 1254 (11.9) | 1186 (11.1) | 1374 (10.9) | 1461 (11.2) | 1576 (11.1) | 1766 (11.3) | 1501 (10.7) |
| 6-12 hour | 1320 (13.4) | 1458 (13.8) | 1419 (13.3) | 1707 (13.5) | 1840 (14.1) | 1963 (13.9) | 2080 (13.3) | 1976 (14.1) |
| 12-24 hour | 1646 (16.8) | 1741 (16.5) | 1824 (17.1) | 2060 (16.3) | 2177 (16.7) | 2441 (17.2) | 2778 (17.8) | 2460 (17.6) |
| 24-36 hour | 721 (7.3) | 751 (7.1) | 788 (7.4) | 971 (7.7) | 869 (6.7) | 1081 (7.6) | 1175 (7.5) | 989 (7.1) |
| 36-48 hour | 498 (5.1) | 495 (4.7) | 518 (4.8) | 688 (5.4) | 625 (4.8) | 715 (5.0) | 750 (4.8) | 661 (4.7) |
| 2-7 day | 1784 (18.2) | 1946 (18.4) | 1845 (17.2) | 2254 (17.8) | 2273 (17.4) | 2381 (16.8) | 2690 (17.2) | 2417 (17.2) |
| >7 day | 295 (3.0) | 337 (3.2) | 368 (3.4) | 439 (3.5) | 457 (3.5) | 462 (3.3) | 443 (2.8) | 418 (3.0) |

Based on 100,586 acute stroke patients

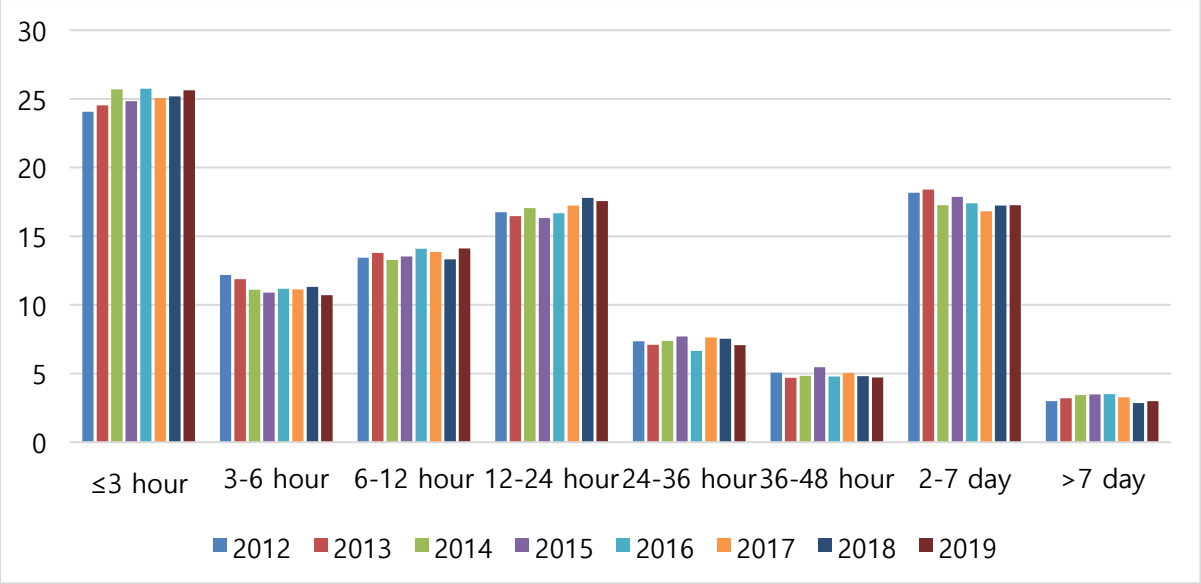


Figure 7. Secular trends of categorized symptom onset to arrival time

Table 8. Anthropometric measurements of acute ischemic stroke and transient ischemic attack patients

| | Mean (SD) | Median (IQR) | Min-Max | Missing (%) | Outliers (%) |
|---|---|---|---|---|---|
| | | | Total | | |
| Height (cm) | 162.4±9.0 | 163 (155 - 170) | 110 - 200 | 2441 (2.5) | 63 (0.1) |
| Weight (kg) | 62.9±11.9 | 62 (55 - 70) | 20 - 181 | 1846 (1.9) | 2 (0.0) |
| BMI (kg/m$^2$) | 23.8±3.5 | 24 (21 - 26) | 11 - 74 | 2657 (2.8) | 65 (0.1) |
| Initial SBP (mmHg) | 148.4±27.7 | 146 (130 - 165) | 50 - 300 | 583 (0.6) | 6 (0.0) |
| Initial DBP (mmHg) | 84.7±15.8 | 82 (74 - 94) | 10 - 220 | 588 (0.6) | 0 (0.0) |
| | | | Female | | |
| Height (cm) | 154.6±6.1 | 155 (150 - 159) | 110 - 190 | 1194 (3.0) | 36 (0.1) |
| Weight (kg) | 56.3±10.1 | 55 (50 - 62) | 20 - 181 | 921 (2.3) | 2 (0.0) |
| BMI (kg/m$^2$) | 23.5±3.8 | 23 (21 - 26) | 11 - 73 | 1302 (3.2) | 38 (0.1) |
| Initial SBP (mmHg) | 148.8±28.5 | 147 (130 - 167) | 50 - 289 | 250 (0.6) | 2 (0.0) |
| Initial DBP (mmHg) | 83.8±15.9 | 80 (72 - 92) | 10 - 210 | 254 (0.6) | 0 (0.0) |
| | | | Male | | |
| Height (cm) | 167.9±6.3 | 168 (164 - 172) | 111 - 200 | 1247 (2.2) | 27 (0.05) |
| Weight (kg) | 67.6±10.8 | 67 (60 - 74) | 22 - 165 | 925 (1.6) | 0 (0.00) |
| BMI (kg/m$^2$) | 23.9±3.2 | 24 (22 - 26) | 11 - 74 | 1355 (2.4) | 27 (0.05) |
| Initial SBP (mmHg) | 148.2±27.1 | 146 (130 - 163) | 50 - 300 | 333 (0.6) | 4 (0.01) |
| Initial DBP (mmHg) | 85.3±15.7 | 83 (75 - 95) | 12 - 220 | 334 (0.6) | 0 (0.00) |

Based on 96,295 acute ischemic stroke and transient ischemic attack patients

Note: The outlier cases are excluded from the analysis in the following cases: height, not between 100-250; weight, not between 20-200; initial systolic BP, not between 50-300; initial diastolic BP, more than 250. BMI:body mass index; SBP: systolic blood pressure; DBP: diastolic blood pressure.

Table 9. Distribution of prestroke mRS score

| Prestroke mRS score | Frequency (n) | Percentage (%) |
|---|---|---|
| 0 | 67936 | 76.1 |
| 1 | 7613 | 8.5 |
| 2 | 5439 | 6.1 |
| 3 | 4653 | 5.2 |
| 4 | 2448 | 2.7 |
| 5 | 1187 | 1.3 |
| Total | 89276 | 100.0 |

Based on 89,276 ischemic stroke cases

Table 10. History of major vascular risk factors and their management before index stroke

| Risk factors and prior managements | Frequency (n) | Percent (%) |
|---|---|---|
| Hypertension | 59509 | 66.7 |
| History of Hypertension | 52971 | 91.4 |
| Diagnosed at the index admission | 5013 | 8.7 |
| Diabetes | 29050 | 32.5 |
| History of Diabetes | 25210 | 87.7 |
| Diagnosed at the index admission | 3525 | 12.3 |
| Dyslipidemia | 32910 | 36.9 |
| History of Dyslipidemia | 17536 | 63.6 |
| Diagnosed at the index admission | 10044 | 36.4 |
| Habitual smoking | 32787 | 36.7 |
| Current smoker | 20527 | 63.6 |
| Ex-smoker, quit within 5 years | 7922 | 24.5 |
| Ex-smoker, quit more than 5 years ago | 3831 | 11.9 |
| Atrial fibrillation | 18021 | 20.5 |
| History of Atrial fibrillation | 9914 | 55.6 |
| Diagnosed at the index admission | 7931 | 44.4 |

Based on 89,277 ischemic stroke cases

* Percentages of management categories were calculated based on the frequency of each risk factor.

Table 11. NIHSS score at hospital arrival

| Region | Mean (SD) | Median (IQR) | Min-Max |
|---|---|---|---|
| Seoul | 5.2±5.6 | 3 (1 - 7) | 0 - 40 |
| Gyeonggi & & Incheon | 5.6±6.3 | 3 (1 - 7) | 0 - 41 |
| Gyeongnam & Busan | 5.9±6.1 | 4 (1 - 8) | 0 - 38 |
| Chungnam & Daejeon | 5.7±6.3 | 4 (1 - 8) | 0 - 38 |
| Gyeongbuk & Daegu | 5.4±6.1 | 3 (1 - 7) | 0 - 37 |
| Gangwon | 4.9±5.6 | 3 (1 - 7) | 0 - 30 |
| Gwangju & Jeonnam | 5.4±5.6 | 3 (1 - 9) | 0 - 31 |
| Chungbuk | 5.7±5.8 | 4 (2 - 8) | 0 - 30 |
| Jeju | 7.0±7.0 | 4 (2 - 10) | 0 - 38 |
| Total | 5.5±6.0 | 3 (1 - 7) | 0 - 41 |

Based on 89,276 ischemic stroke cases

NIHSS: National Institutes of Health Stroke Scale

Table 12. Profile of method of revascularization therapies for ischemic strokes

| Region | IVT | IAT | Combined | Total |
|---|---|---|---|---|
| Seoul | 1519 (6.5) | 594 (2.5) | 571 (2.4) | 2684 (11.4) |
| Gyeonggi & & Incheon | 1528 (7.2) | 919 (4.3) | 822 (3.9) | 3269 (15.4) |
| Gyeongnam & Busan | 1545 (10.3) | 713 (4.8) | 749 (5.0) | 3007 (20.1) |
| Chungnam & Daejeon | 318 (9.5) | 56 (1.7) | 101 (3.0) | 475 (14.2) |
| Gyeongbuk & Daegu | 664 (5.9) | 618 (5.5) | 384 (3.4) | 1666 (14.9) |
| Gangwon | 177 (8.1) | 46 (2.1) | 68 (3.1) | 291 (13.3) |
| Gwangju & Jeonnam | 876 (10.8) | 395 (4.9) | 356 (4.4) | 1627 (20.0) |
| Chungbuk | 117 (6.3) | 115 (6.1) | 113 (6.0) | 345 (18.4) |
| Jeju | 260 (12.0) | 88 (4.1) | 120 (5.5) | 468 (21.6) |
| Total | 7004 (7.9) | 3544 (4.0) | 3284 (3.7) | 13832 (15.6) |

Based on 88,669 ischemic stroke cases

IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

Table 13. Secular trends of revascularization modalities

| | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | Total |
|---|---|---|---|---|---|---|---|---|---|
| IVT | 721 (8.4) | 800 (8.6) | 908 (9.7) | 872 (8.1) | 892 (7.9) | 896 (7.2) | 982 (6.9) | 933 (7.3) | 7004 (7.9) |
| IAT | 210 (2.5) | 215 (2.3) | 270 (2.9) | 339 (3.2) | 425 (3.8) | 519 (4.2) | 839 (5.9) | 727 (5.7) | 3544 (4.0) |
| Combined | 248 (2.9) | 279 (3.0) | 369 (3.9) | 395 (3.7) | 421 (3.7) | 484 (3.9) | 557 (3.9) | 531 (4.2) | 3284 (3.7) |

Based on 88,669 ischemic stroke cases

IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

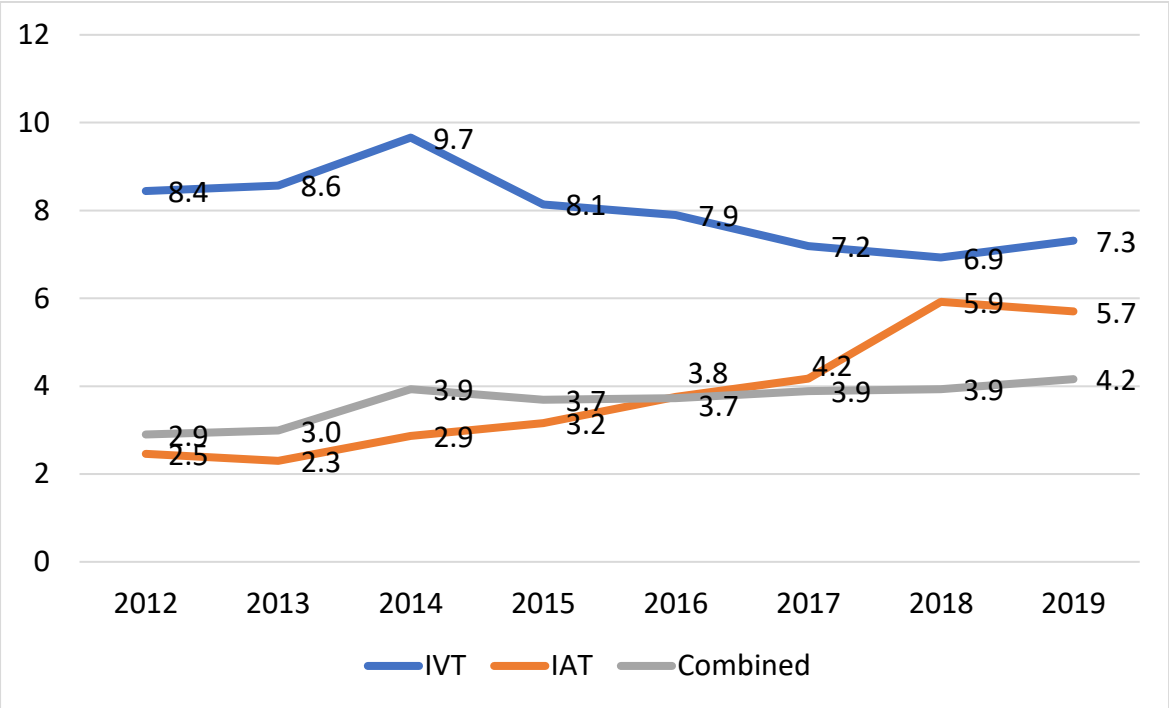


Figure 8. Secular trends of revascularization modalities

Based on 88,669 ischemic stroke cases

IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

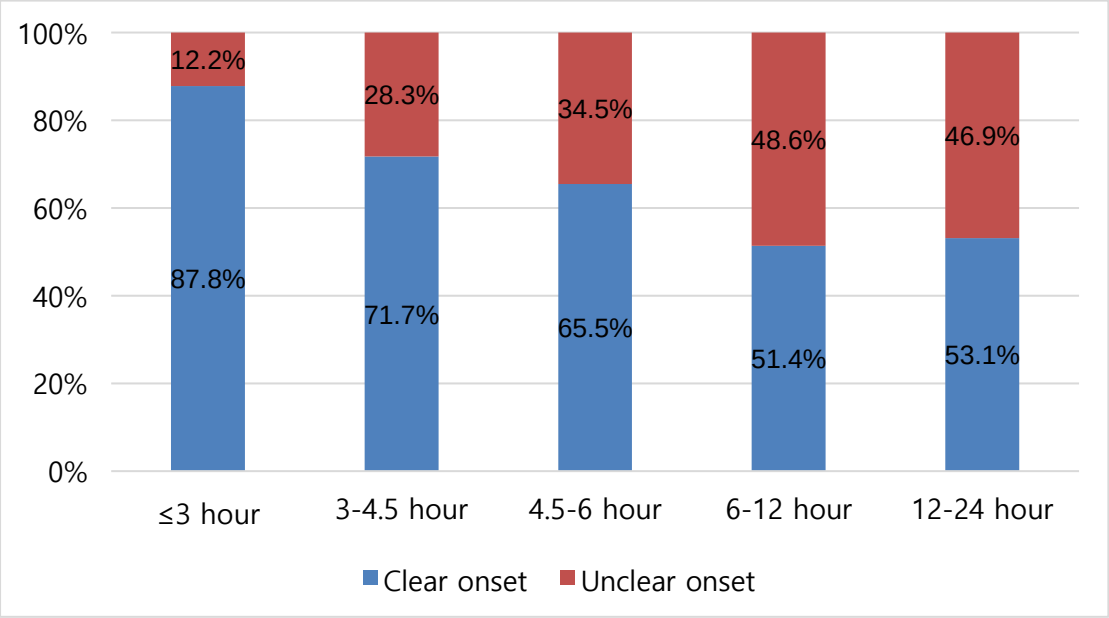


Figure 9. Proportions of clear onset, unclear onset by categorized onset (LNT) to arrival time

Based on 58,422 ischemic stroke cases

Table 14. Profile of laboratory tests

| Laboratory variables | N | Mean ± SD | Median [IQR] | Min-Max |
|---|---|---|---|---|
| Blood Urea Nitrogen (mg/dL) | 88619 | 17.53±9.11 | 16 (12 - 20) | 0.04 - 206 |
| Creatinine (mg/dL) | 88618 | 1.01±1.12 | 1 (1 - 1) | 0.01 - 116 |
| Hemoglobin (g/dL) | 83038 | 13.59±2.04 | 14 (12 - 15) | 0.10 - 50 |
| TG | 86321 | 125.60±88.91 | 104 (75 - 148) | 0.80 - 4970 |
| LDL-cholesterol (mg/dL) | 86133 | 106.87±37.74 | 104 (80 - 130) | 0.30 - 640 |
| HDL-cholesterol (mg/dL) | 85007 | 45.37±13.68 | 44 (36 - 52) | 3.00 - 664 |
| Platelet count ($x10^3$µL) | 83001 | 230.63±73.32 | 223 (184 - 268) | 5.33 - 990 |
| Total cholesterol (mg/dL) | 87035 | 172.59±44.80 | 170 (142 - 200) | 6.00 - 800 |
| Initial Glucose (mg/dL) | 82261 | 145.35±64.40 | 126 (106 - 161) | 0.03 - 2111 |
| Fasting Glucose (mg/dL) | 80907 | 123.11±50.30 | 109 (94 - 136) | 0.60 - 1042 |
| Hb A1c (%) | 75883 | 6.35±2.04 | 6 (6 - 7) | 0.05 - 209 |
| SBP (mm Hg) | 88743 | 148.77±27.84 | 147 (130 - 165) | 13 - 300 |
| DBP (mm Hg) | 88739 | 84.82±15.87 | 82 (74 - 94) | 10 - 220 |

Based on 89,227 ischemic stroke cases

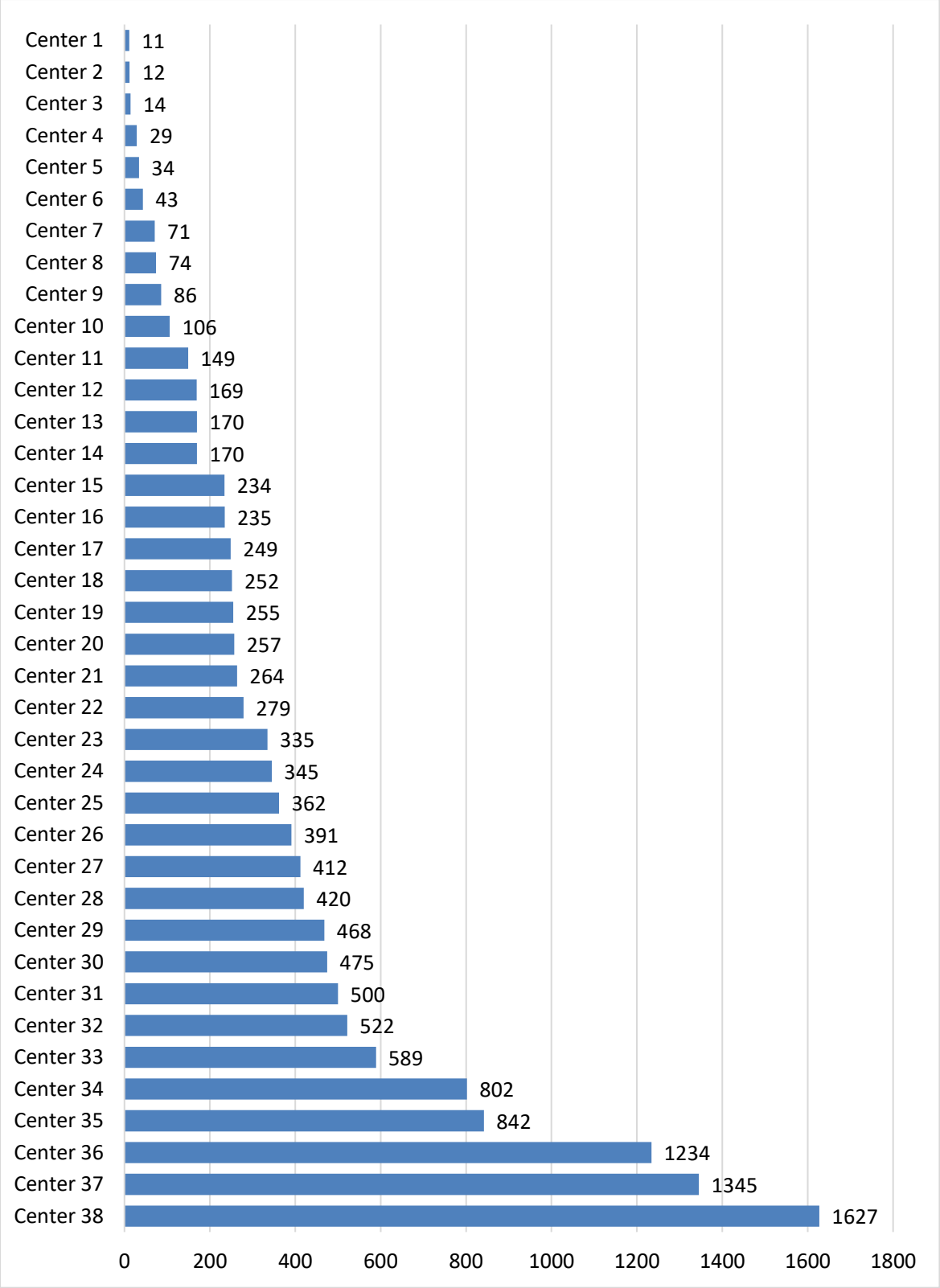


Figure 10. Volume of acute revascularization treatment for each stroke center (Anonymized)

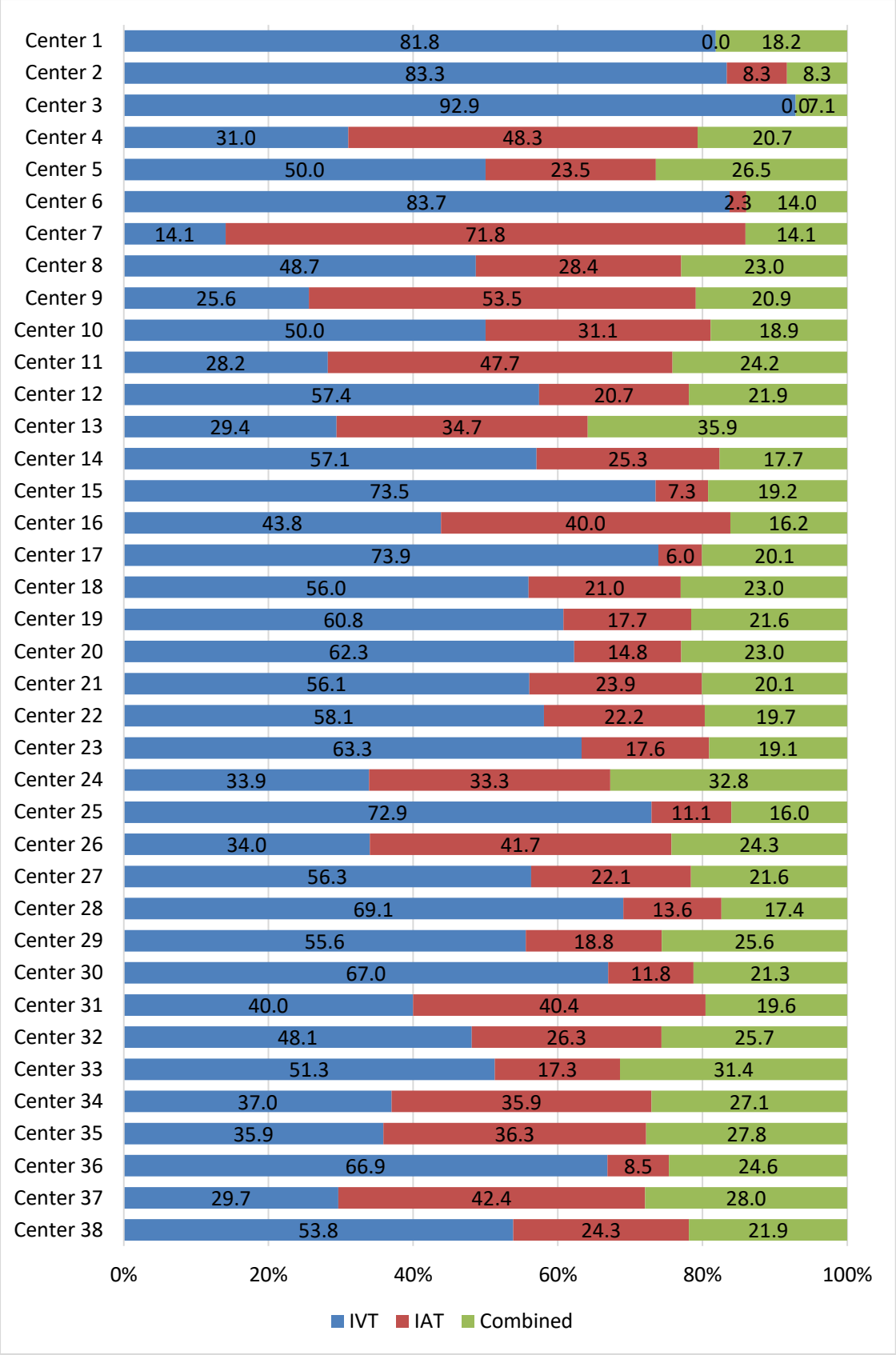


Figure 11. Discrepancy in revascularization modalities among centers

IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

Table 15. Regional disparity of time interval between symptom onset and hospital arrival

| | Percent of patients, % | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Seoul | Gyeonggi & Incheon | Gyeongnam & Busan | Chungnam & Daejeon | Gyeongbuk & Daegu | Gangwon | Gwangju & Jeonnam | Chungbuk | Jeju |
| ≤3 hour | 22.2 | 24.8 | 27.8 | 27.8 | 23.1 | 26.6 | 27.5 | 31.7 | 30.1 |
| 3-6 hour | 9.3 | 10.2 | 13.0 | 9.6 | 11.4 | 10.9 | 17.2 | 10.6 | 11.5 |
| 6-12 hour | 13.1 | 13.0 | 14.6 | 14.1 | 13.5 | 12.6 | 15.3 | 14.2 | 14.8 |
| 12-24 hour | 17.2 | 17.1 | 16.9 | 17.4 | 18.1 | 17.7 | 14.4 | 18.0 | 16.8 |
| 24-36 hour | 7.9 | 7.1 | 6.5 | 8.1 | 7.1 | 7.2 | 8.5 | 5.0 | 6.4 |
| 36-48 hour | 5.9 | 5.1 | 4.0 | 5.1 | 5.4 | 4.8 | 2.8 | 6.2 | 3.5 |
| 2-7 day | 20.8 | 18.7 | 14.5 | 16.3 | 17.8 | 17.0 | 12.1 | 14.0 | 15.7 |
| >7 day | 3.6 | 3.9 | 2.7 | 1.6 | 3.7 | 3.3 | 2.3 | 0.2 | 1.3 |

Based on 100,586 ischemic stroke cases

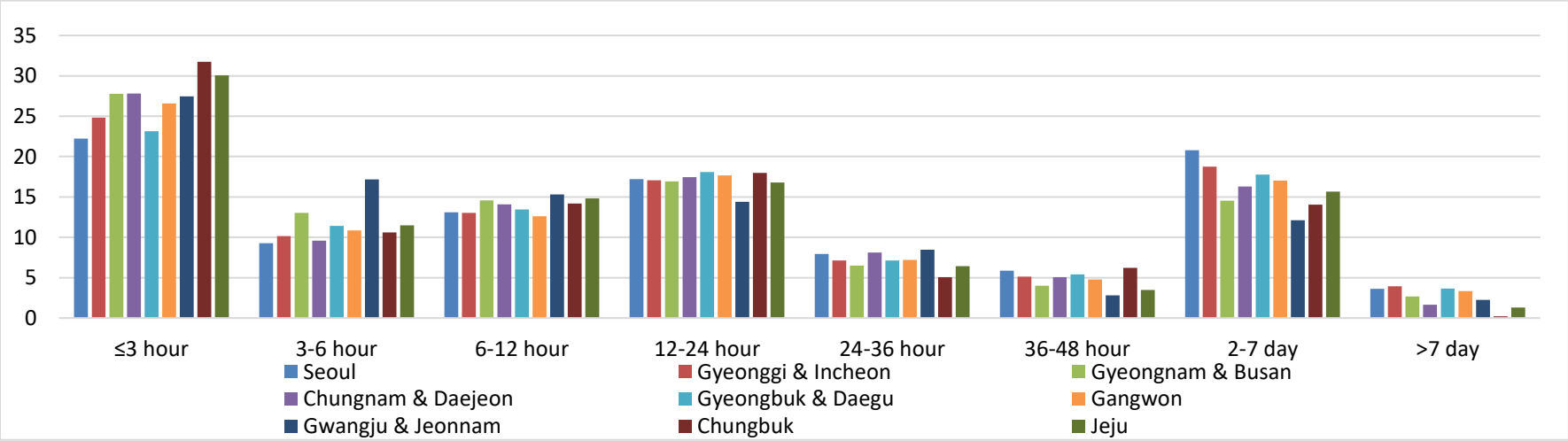


Figure 12. Regional disparity of time interval between symptom onset and hospital arrival

Table 16. Distribution and secular trends of mRS score at discharge

| | Number of patients (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| mRS | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
| 0 | 1247 (15.0) | 1259 (13.8) | 1467 (15.9) | 1715 (15.9) | 1971 (17.6) | 1894 (15.4) | 2065 (14.6) | 1860 (14.5) |
| 1 | 2037 (24.5) | 2113 (23.2) | 2237 (24.3) | 2641 (24.5) | 2716 (24.2) | 3150 (25.6) | 3626 (25.6) | 3672 (28.7) |
| 2 | 1544 (18.6) | 1927 (21.2) | 1779 (19.3) | 2307 (21.4) | 2199 (19.6) | 2471 (20.1) | 2652 (18.7) | 2245 (17.5) |
| 3 | 1280 (15.4) | 1454 (16.0) | 1342 (14.6) | 1607 (14.9) | 1617 (14.4) | 1809 (14.7) | 2306 (16.3) | 1957 (15.3) |
| 4 | 1183 (14.2) | 1268 (13.9) | 1308 (14.2) | 1412 (13.1) | 1512 (13.5) | 1616 (13.1) | 1818 (12.8) | 1710 (13.4) |
| 5 | 802 (9.7) | 837 (9.2) | 834 (9.1) | 837 (7.8) | 928 (8.3) | 1120 (9.1) | 1349 (9.5) | 1075 (8.4) |
| 6 | 216 (2.6) | 239 (2.6) | 248 (2.7) | 239 (2.2) | 283 (2.5) | 263 (2.1) | 350 (2.5) | 285 (2.2) |

Based on 87,898 ischemic stroke cases

mRS: modified Rankin Scale

# KSR Annual Report

## *2020*

Korean Stroke Registry
한국뇌졸중등록사업

대한뇌졸중학회
Korean Stroke Society

Korea Disease Control and Prevention Agency
National
Institute of Health